

BLUESTOCK MUTUAL FUND ANALYTICS DASHBOARD

Final Project Report

Data Analyst Internship Program

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Executive Summary

This report documents the work carried out during my Data Analyst Internship at Bluestock Fintech, where I was assigned to build the Bluestock Mutual Fund Analytics Dashboard. The project was designed to give internal stakeholders and prospective investors a single, consolidated view of mutual fund industry data that would otherwise sit scattered across separate spreadsheets and scheme-level reports. Over the course of the internship, I worked through the full analytics lifecycle: understanding the raw data, cleaning and structuring it, designing a data model, and finally building an interactive Power BI dashboard that presents industry trends, fund performance, investor behavior, and SIP activity in one place.

The purpose of the dashboard is to reduce the manual effort involved in comparing mutual fund schemes and to make risk, return, and investor activity patterns easier to interpret at a glance. Before a tool like this exists, someone trying to answer a simple question such as "which fund house has the largest share of industry assets" or "how are SIP inflows trending against the market" would need to pull numbers from multiple sources and reconcile them manually. The dashboard addresses this by pulling the relevant metrics into four dedicated pages, each built around a specific business question.

The underlying business problem is that mutual fund data, while publicly available in various forms, is not naturally presented in a way that supports quick decision-making. Investors and internal analysts need to compare funds not just on returns, but on risk-adjusted performance, and they need to understand how retail investor behavior, lumpsum versus SIP, participation by age group.

From a workflow perspective, the project followed a standard ETL approach. Raw data was extracted from source files and an application database, cleaned and validated during a transformation stage, and then loaded into a data warehouse layer before being connected to Power BI for modeling and visualization. This structure, captured in the architecture diagram included later in this report, kept the data preparation work separate from the reporting layer, which made the dashboard easier to maintain and extend.

The major outcome of the project is a four-page interactive dashboard covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. Each page uses filters and slicers so that a user can narrow the view to a specific fund house, category, plan, state, age group, or time period without needing any technical knowledge of the underlying data. Key metrics such as total AUM, SIP inflows, number of folios, active schemes, risk-adjusted return ratios, and transaction patterns are all made visible through this interface.

Overall, this project gave me practical exposure to what it actually takes to turn raw financial data into something decision-makers can use. It reinforced the importance of getting the data model right before touching visuals, and it showed me how much value a well-organized dashboard can add for an organization operating in the mutual fund space.

Project Overview

Background

Mutual fund analytics has become increasingly important as retail participation in mutual funds continues to grow. Fund houses, distributors, and investors alike depend on accurate and timely data to make decisions about where money should be allocated. However, mutual fund data is inherently multidimensional, it spans asset classes, categories, plans, time periods, and investor segments. Which makes it difficult to interpret in raw tabular form. Effective visualization is what allows this complexity to be reduced to a set of clear, comparable metrics. A well-designed dashboard does not just display numbers; it organizes them so that patterns such as concentration of assets, risk-return trade-offs, and shifts in investor behavior become visible without requiring the viewer to run their own analysis each time.

Problem Statement

Stakeholders looking at mutual fund data typically face three related challenges. First, the sheer volume of data like spanning multiple fund houses, dozens of schemes, and thousands of investor transactions, makes manual review impractical. Second, comparing funds fairly requires looking at more than just returns; risk grade, Sharpe ratio, and Sortino ratio all need to be considered together, and calculating or locating these consistently across funds is time-consuming without a dedicated tool. Third, understanding investor behavior, how contributions are split between lumpsum and SIP, which regions and age groups are most active, and how monthly transaction volumes are trending, requires joining transaction-level data with demographic information, which is not something a static report handles well.

Project Objectives

The project was structured around five core objectives. The first was industry trend analysis, which involved tracking overall assets under management, folio counts, and scheme counts over time, and comparing asset size across the leading asset management companies. The second was fund performance analysis, which meant building the tools to compare funds on both raw returns and risk-adjusted metrics, so that a fund's performance could be judged relative to the risk it carries rather than in isolation. The third objective was investor behavior analysis, aimed at understanding

how transaction value, transaction type, and SIP participation vary by state, age group, and city tier. The fourth was SIP and market trend analysis, which focused on how SIP inflows have moved over time relative to a broad market benchmark, and which fund categories have attracted the most net inflow in the most recent financial year. The fifth and final objective was to bring all of this together into a single interactive dashboard in Power BI, with filtering and drill-down capability, so that the analysis would remain usable by non-technical stakeholders rather than being locked inside a one-off report.

Data Sources

The dashboard is built on four related datasets, each exported in CSV format and later cleaned in Excel before being loaded into the analytics workflow. Working with CSV files kept the data source lightweight and easy to version during development, though it also meant that a fair amount of care was needed during the transformation stage to keep formats, category names, and fund house names consistent across files. Mismatch as small as an extra space in a fund house name is enough to break a relationship in the data model.

Each dataset corresponds to one of the analytical objectives described in the previous section: an industry-level dataset covering AMC and scheme figures, a fund performance dataset covering returns and risk metrics, an investor transaction dataset covering transaction-level activity, and a SIP and market trend dataset covering inflow figures against a market benchmark. The table below summarizes each dataset.

Dataset Name	Description	Purpose
Industry & AMC Data	Scheme-wise and AMC-wise asset figures, folio counts and active scheme counts covering FY2022–2025.	Industry trend analysis
Fund Performance Data	Fund-wise NAV history, 1-year and 3-year returns, risk grade, Sharpe ratio, Sortino ratio and AUM, segmented by category and plan.	Fund performance and risk analysis
Investor Transaction Data	Transaction-level records tagged by state, age group, city tier and transaction type (lumpsum, SIP, redemption).	Investor behavior analysis
SIP & Market Trend Data	Monthly and yearly SIP inflow figures by category and fund house, alongside Nifty 50 index movement.	SIP and market trend analysis

Data quality was an ongoing consideration throughout the project. Because the datasets originate from different reporting structures, category labels and fund house names needed to be standardized before they could be joined reliably, and numeric fields required consistent formatting so that aggregations in Power BI would compute correctly. Missing or incomplete records were reviewed rather than silently dropped, so that any gaps in the data would not be mistaken for zero-value activity.

ETL Design

The dashboard is supported by an ETL (Extract, Transform, Load) pipeline that separates raw data collection from the reporting layer. The diagram below, prepared during the design phase of the project, illustrates how data moves from the source application through to the final business insights delivered in Power BI.

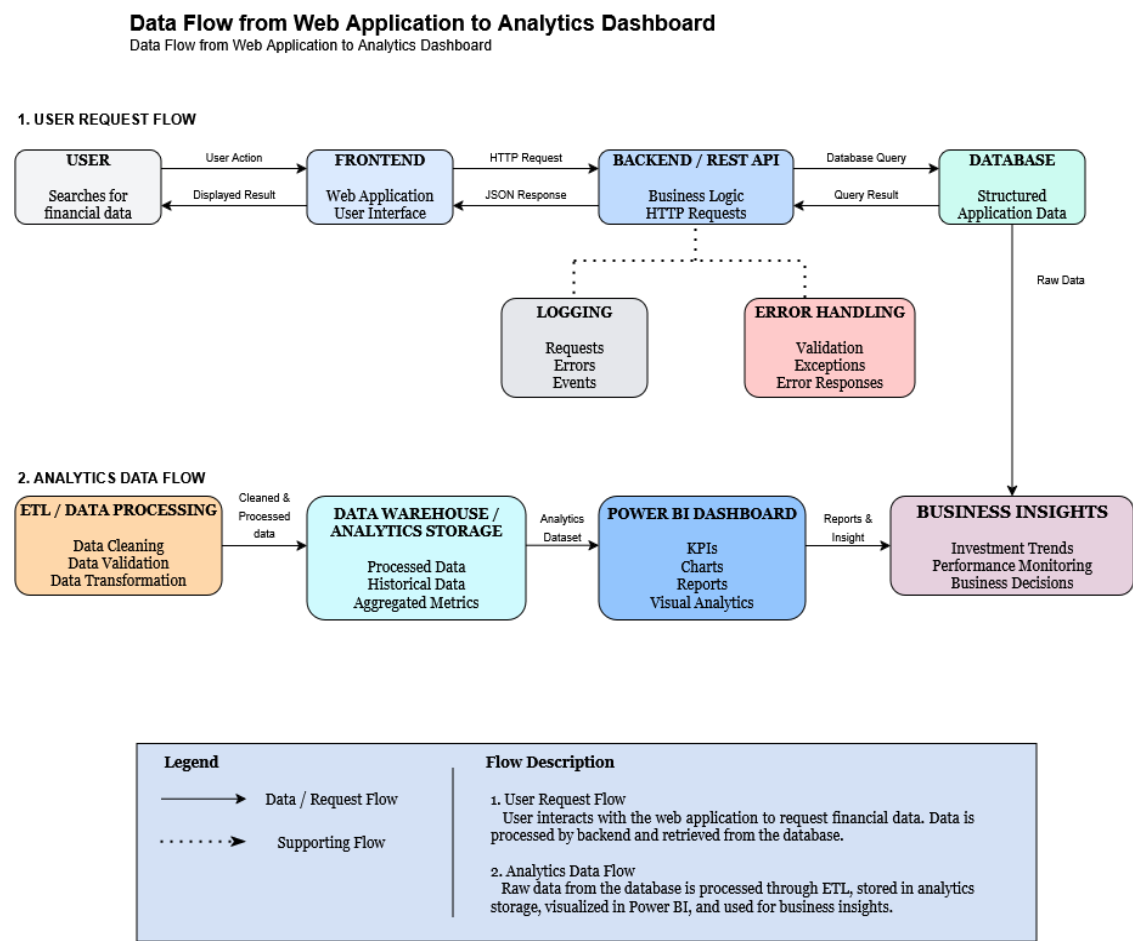


Figure 1: Data flow architecture from the web application to the analytics dashboard.

Extract

Data originates from user activity on the web application. As shown in the upper half of the diagram, a user action triggers an HTTP request from the frontend to the backend REST API, which in turn queries the underlying database and returns structured application data. Logging and error handling run alongside this request flow, capturing request activity, exceptions and validation

errors so that data quality issues can be traced back to their source rather than surfacing only once they reach the dashboard. This raw, structured data is the extraction point for the analytics pipeline.

Transform

The lower half of the diagram shows the analytics data flow, which begins with an ETL / Data Processing stage. At this stage, raw data pulled from the database goes through data cleaning, validation and transformation before it is allowed into the analytics layer. In practical terms, this meant standardizing category and fund house naming conventions across the different source files, checking numeric fields for consistency, and validating that transaction and performance records fell within reasonable ranges before they were accepted into the model. Only cleaned and processed data is passed forward, which is what keeps the dashboard's figures reliable.

Load

Once cleaned, the data is loaded into a data warehouse/analytics storage layer, which holds processed data, historical data and aggregated metrics. From there, the analytics dataset is loaded into the Power BI dashboard, where it is modeled into the KPIs, charts and reports that make up the four dashboard pages. The final step in the diagram, feeding reports and insight back into business decisions reflects the actual purpose of the pipeline: the ETL process exists to support investment trend analysis, performance monitoring and business decision-making, not simply to move data from one place to another.

Exploratory Data Analysis

This section works through the two dashboard pages that form the analytical foundation of the project Industry Overview and Fund Performance, describing what each visualization shows and what can reasonably be concluded from it.

Industry Overview Dashboard



Figure 2: Industry Overview dashboard page.

Visualization Description: This page opens with four KPI cards — Total AUM (₹10.44L Cr), SIP Inflows (₹9.40K Cr), Folios (5K) and Schemes (40) — followed by an Industry AUM Trend chart spanning FY2022 to FY2025 and an AUM by AMC bar chart ranking the top five fund houses by asset size.

Observations: The Industry AUM Trend chart holds in a fairly narrow band across the FY2022–2025 period shown, without the sharp jumps or dips that would suggest a volatile industry. The AUM by AMC chart shows Nippon India MF at the top, followed closely by Kotak Mahindra MF, then ICICI Prudential MF, DSP Mutual Fund and Aditya Birla Sun Life MF, the five bars sit fairly close together rather than showing one dominant outlier.

Insights: The closeness of the top five AMCs by asset size suggests a competitive landscape at the top of the industry rather than a single dominant player. With SIP inflows at ₹9.40K Cr against

a total AUM of ₹10.44L Cr, systematic investment plans clearly represent a meaningful and recurring contributor to industry assets rather than a marginal channel.

Interpretation: For a fintech platform like Bluestock, this view matters because it frames where the platform's own fund offerings sit relative to established players, and it signals that SIP-based products are worth prioritizing given how consistently they are shown to feed into industry-wide asset growth.

Fund Performance Dashboard

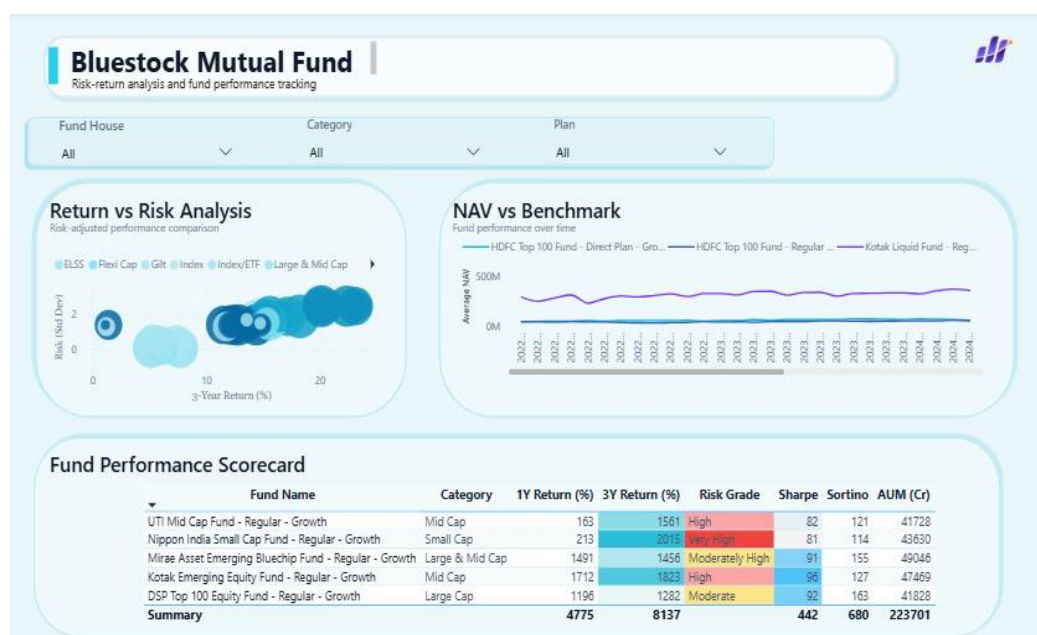


Figure 3: Fund Performance dashboard page.

Visualization Description: This page provides Fund House, Category and Plan filters above two charts, a Return vs Risk Analysis bubble chart plotting 3-Year Return against Risk (standard deviation) by category, and a NAV vs Benchmark line chart tracking average NAV over time for selected funds. Followed by a Fund Performance Scorecard table listing individual funds with their category, 1-year and 3-year returns, risk grade, Sharpe ratio, Sortino ratio and AUM.

Observations: In the bubble chart, most category bubbles cluster toward the lower end of the risk axis with modest returns, while at least one category sits further out on both axes, indicating a segment with materially higher risk and higher potential return. The scorecard shows a wide spread across the five listed funds: for example, Nippon India Small Cap Fund and UTI Mid Cap

Fund carry High or Very High risk grades alongside noticeably higher 3-year returns, while DSP Top 100 Equity Fund shows a more moderate risk grade with a comparatively lower 3-year return.

Insights: Return and risk grade move together in the scorecard more often than not, which is broadly consistent with financial theory, higher return is generally accompanied by higher volatility. The presence of both high-risk, high-return funds and lower-risk, steadier funds in the same view means the dashboard can support very different investor profiles from a single page.

Interpretation: This matters because it shifts the comparison away from returns alone and toward risk-adjusted performance, which is a more defensible basis for recommending a fund to an investor with a specific risk appetite.

Performance Analysis

This section takes a closer look at the fund-level metrics captured in the Fund Performance Scorecard, treating them as a performance review rather than simply describing the dashboard again.

Metric Explanation

The scorecard tracks six metrics per fund. The 1-Year and 3-Year Return figures measure the fund's price appreciation over those respective periods. Risk Grade is a categorical rating (Moderate, Moderately High, High, Very High in the data shown) that summarizes the fund's volatility relative to its category. The Sharpe ratio measures return earned per unit of total risk taken, while the Sortino ratio is a related measure that focuses specifically on downside risk rather than total volatility. AUM (in crores) indicates the scale of assets the fund manages, which is a rough proxy for investor confidence and liquidity.

Analysis

Looking at the five funds in the scorecard, Nippon India Small Cap Fund shows the highest 3-year return among the listed funds at 2015 percent, but it also carries a Very High risk grade — the highest in the table. UTI Mid Cap Fund follows a similar pattern with a High risk grade and a strong 3-year return of 1561 percent. By contrast, DSP Top 100 Equity Fund shows the most moderate risk grade in the group alongside the lowest 3-year return of the five funds at 1282 percent, while Kotak Emerging Equity Fund and Mirae Asset Emerging Bluechip Fund sit in between on both risk and return. Sharpe ratios across the five funds range from 82 to 96, and Sortino ratios range from 114 to 163, with Kotak Emerging Equity Fund showing the highest Sharpe ratio (91) and Sortino ratio (127) relative to its risk grade of High — suggesting comparatively efficient risk-adjusted performance within this set.

Business Interpretation

For an investor prioritizing growth and willing to accept volatility, the small-cap and mid-cap funds in this scorecard would be the more relevant options given their stronger returns, provided the accompanying High or Very High risk grade is acceptable to them. For a more conservative investor, DSP Top 100 Equity Fund's moderate risk grade makes it a more defensible

recommendation even though its raw return trails the others in this table. This is exactly the kind of trade-off the dashboard is designed to surface — it does not tell the user which fund is "best" in the abstract, but it puts return and risk side by side so the decision can be made against a specific risk tolerance.

Key Takeaway

Raw return figures alone are not a sufficient basis for fund comparison. The scorecard's inclusion of risk grade alongside Sharpe and Sortino ratios is what allows a genuinely risk-adjusted comparison, and this is the core value the Fund Performance page adds over simply listing returns.

Dashboard Overview

The dashboard is organized into four pages, each addressing one of the project's core objectives. This section walks through each page individually, covering its purpose, the components it contains, how a user interacts with it, and the business value it provides.

Page 1 — Industry Overview

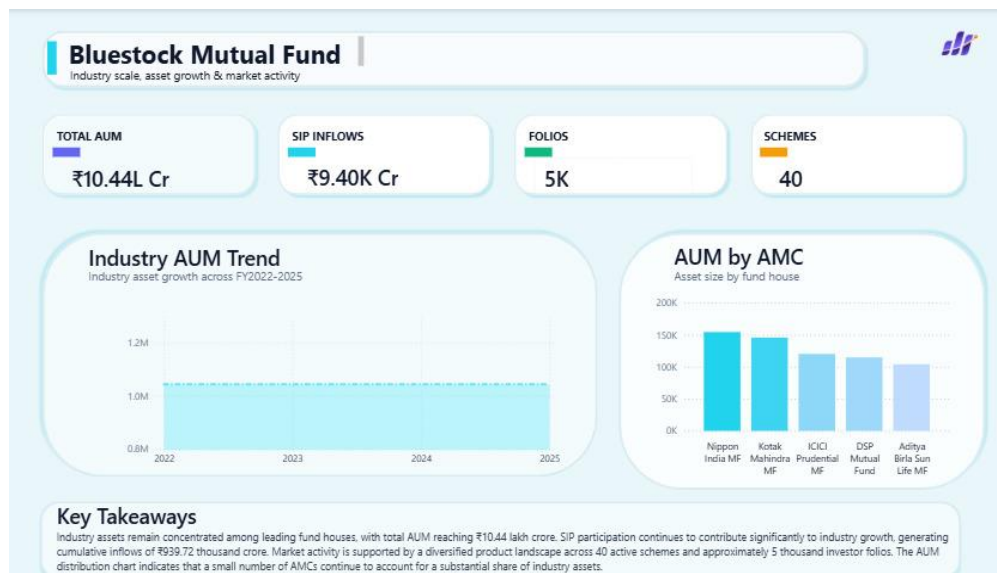


Figure 4: Industry Overview page.

Purpose: This page exists to give a top-level snapshot of the mutual fund industry before a user drills into any single fund. It answers the question "how large and how concentrated is the industry right now" in one view.

Components: Four KPI cards — Total AUM, SIP Inflows, Folios and Schemes — sit at the top of the page, followed by an Industry AUM Trend chart covering FY2022–2025 and an AUM by AMC bar chart comparing the top five asset management companies by size. A Key Takeaways text box beneath the charts summarizes the page's main points in plain language.

User Interaction: This page is presented as a fixed summary view rather than a heavily filterable one, which is appropriate for its role as a starting point — a user is meant to absorb the industry-level picture here before moving to the more detailed, filterable pages that follow.

Business Value: By putting total industry scale and AMC concentration on the very first page, this view orients any stakeholder — whether an internal analyst or a prospective investor — before they engage with more granular fund-level or investor-level data on the pages that follow.

Page 2 — Fund Performance

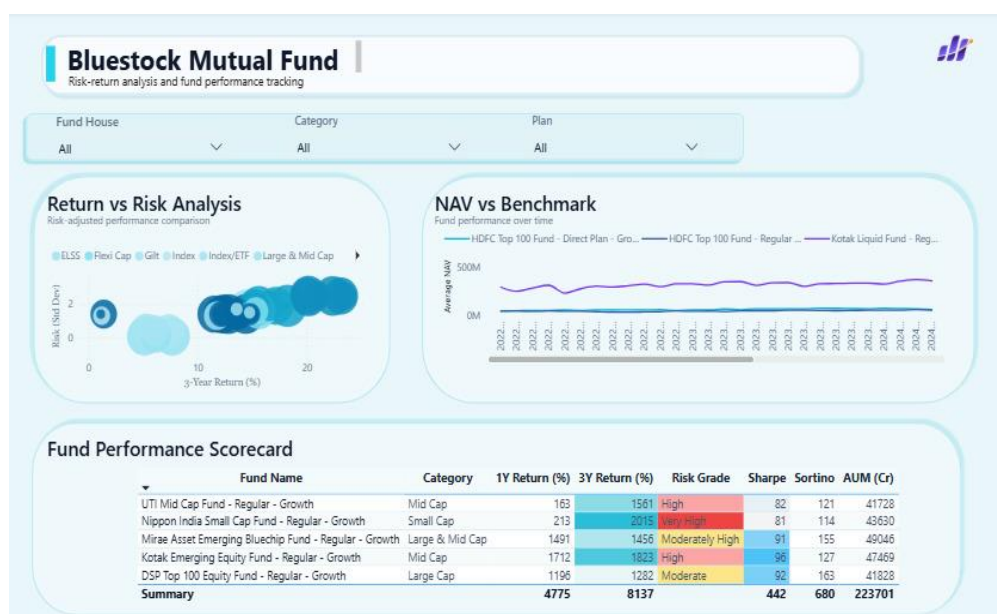


Figure 5: Fund Performance page.

Purpose: This page supports direct comparison between funds on a risk-adjusted basis, which is the central analytical task for anyone deciding between mutual fund schemes.

Components: Three slicers — Fund House, Category and Plan — sit above a Return vs Risk Analysis bubble chart, a NAV vs Benchmark line chart, and a Fund Performance Scorecard table listing return, risk grade, Sharpe ratio, Sortino ratio and AUM for individual funds.

User Interaction: The Fund House, Category and Plan dropdown filters allow a user to narrow the bubble chart, NAV comparison and scorecard down to a specific segment — for instance, viewing only Large Cap funds on the Direct plan — without needing to rebuild any chart manually.

Business Value: This page turns fund selection from a returns-only comparison into a risk-aware one, which is more defensible when recommending a fund to an investor with a specific risk profile.

Page 3 — Investor Analytics

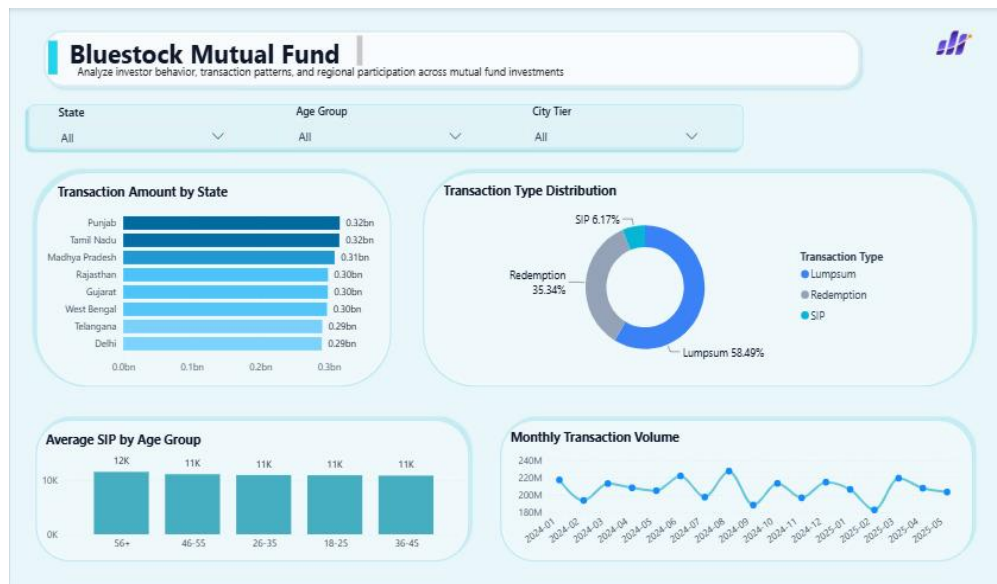


Figure 6: Investor Analytics page.

Purpose: This page shifts the focus from funds to investors, answering questions about who is investing, how, and where, which is essential for shaping distribution and marketing decisions.

Components: State, Age Group and City Tier slicers sit above four visuals — a Transaction Amount by State bar chart, a Transaction Type Distribution donut chart (Lumpsum, SIP, Redemption), an Average SIP by Age Group bar chart, and a Monthly Transaction Volume line chart running from early 2024 through mid-2025.

User Interaction: A user can filter the entire page by state, age group or city tier, letting them isolate, for example, transaction behavior among a specific age bracket in a specific region, with all four visuals updating together.

Business Value: Understanding that Lumpsum transactions make up the majority of transaction value (58.49 percent) compared to SIP (6.17 percent) and Redemption (35.34 percent) in the data shown, alongside which states and age groups drive the highest transaction amounts, gives the business a factual basis for deciding where to focus investor outreach and product design.

Page 4 — SIP & Market Trends

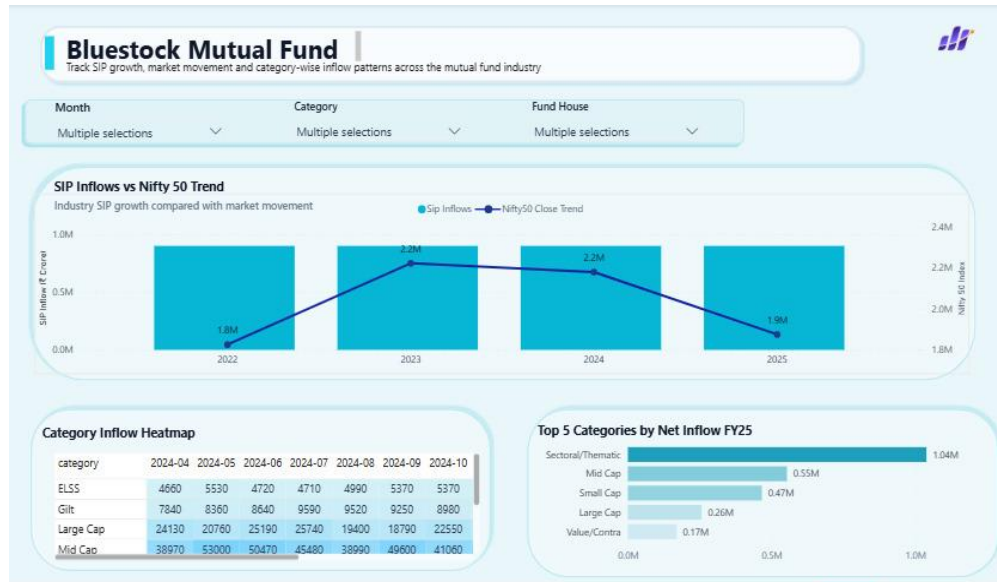


Figure 7: SIP & Market Trends page.

Purpose: This page tracks how SIP activity is moving over time and relative to the broader market, and identifies which fund categories are currently attracting the most net inflow.

Components: Month, Category and Fund House multi-select filters sit above a SIP Inflows vs Nifty 50 Trend combination chart (bars for SIP inflows, a line for the Nifty 50 index), a Category Inflow Heatmap table breaking inflows down by category and month, and a Top 5 Categories by Net Inflow FY25 bar chart.

User Interaction: The multi-select filters allow a user to view several months, categories or fund houses at once rather than being restricted to one selection at a time, which suits the exploratory, trend-comparison nature of this page.

Business Value: Seeing that Sectoral/Thematic funds lead net inflow for FY25 in the data shown, well ahead of Mid Cap and Small Cap, gives the business an early signal of where investor appetite is currently concentrated, which can inform which categories to promote or where additional fund options might be worth developing.

Key Findings

Drawing together the analysis from the previous sections, the following points summarize what the dashboard's data shows across the mutual fund industry, fund performance, investor behavior and SIP activity.

Industry Trends

- Total industry AUM stands at ₹10.44L Cr, spread across 40 active schemes and roughly 5 thousand investor folios.
- Asset size at the top of the industry is fairly evenly distributed among the top five AMCs, led by Nippon India MF, rather than concentrated in a single dominant fund house.

Fund Performance Insights

- Funds carrying higher risk grades (High, Very High) in the scorecard — largely Small Cap and Mid Cap categories — also show the strongest 1-year and 3-year returns among the funds reviewed.
- Risk-adjusted metrics (Sharpe, Sortino) vary meaningfully even among funds with similar risk grades, reinforcing that return alone is not a reliable basis for comparison.

Investor Behavior Insights

- Lumpsum investments account for the majority of transaction value (58.49 percent), well ahead of Redemption (35.34 percent) and SIP (6.17 percent) by transaction count share shown on the dashboard.
- Average SIP contribution is fairly consistent across age groups, with the 56+ age group showing the highest average SIP value (₹12K) among the groups shown.
- Transaction amount by state is led by Punjab and Tamil Nadu, with the remaining states shown clustered closely behind rather than trailing far off.

SIP Trends

- SIP inflows rose from 2022 to a peak around 2023–2024 before easing off in 2025 in the trend shown on the SIP & Market Trends page.

- Sectoral/Thematic funds lead net inflow for FY25, followed by Mid Cap and Small Cap, indicating a current tilt in investor appetite toward higher-growth categories.

Limitations

As with any analytics project built on a fixed data snapshot, this dashboard has a number of limitations worth stating clearly.

- **Historical Data Dependency:** The dashboard reflects data available up to the most recent export used during development. Trends shown, particularly on the Industry AUM Trend and SIP Inflows vs Nifty 50 charts, will need to be refreshed periodically to remain current.
- **Dataset Scope Limitations:** The datasets used cover a defined set of fund houses, categories and time periods. Fund houses or schemes outside this scope are not represented, so industry-wide conclusions should be read as directional rather than exhaustive.
- **Market Volatility Considerations:** Return and risk figures are calculated over fixed historical windows (1-year and 3-year). Market conditions can shift quickly, and figures that were accurate at the time of data extraction may not hold under changed market conditions.
- **Data Availability Constraints:** Some dimensions, such as more granular investor demographics or scheme-level expense ratios, were not available in the source data and are therefore not reflected in the current dashboard.

Recommendations

Based on the work completed, the following enhancements would extend the dashboard's usefulness going forward.

- **Real-Time Data Integration:** Connecting the dashboard to a live or near-live data feed rather than periodic CSV exports would keep the Industry Overview and SIP trend figures current without manual refresh cycles.
- **Predictive Analytics:** Adding forecasting for metrics such as SIP inflows or AUM growth would extend the dashboard from a descriptive tool into a forward-looking one, helping the business anticipate rather than only react to trends.
- **Advanced Investor Segmentation:** Building on the existing State, Age Group and City Tier slicers with more detailed segmentation (for example, by investment tenure or product preference) would sharpen targeting for outreach and product design.
- **Automated Alerts:** Configuring threshold-based alerts — for instance, flagging a sharp month-over-month drop in SIP inflows or a fund crossing into a higher risk grade — would let stakeholders react to changes without needing to check the dashboard manually.
- **Additional Benchmarking Capabilities:** Expanding the NAV vs Benchmark comparison to include more benchmark indices beyond Nifty 50 would give a fuller picture of relative fund performance across different market segments.

Conclusion

This project set out to bring together industry, fund performance, investor behavior and SIP data into a single interactive dashboard, and by the end of the internship period that objective was met. Working through the full pipeline — from understanding the raw CSV sources, through cleaning and structuring the data, to building the Power BI data model and finally the four dashboard pages — gave me a much more grounded understanding of what an analytics project actually involves than any classroom exercise had before.

The work completed covers the five objectives set out at the start of the project: industry trend analysis through the Industry Overview page, fund performance analysis through the risk-return and scorecard visuals on the Fund Performance page, investor behavior analysis through the Investor Analytics page, SIP trend analysis through the SIP & Market Trends page, and the interactive dashboard itself, tying all four together with consistent filtering and a shared visual style.

The major outcomes are a functioning, filterable four-page dashboard and the underlying ETL pipeline that supports it, along with the analytical findings summarized earlier in this report — particularly around the concentration of industry assets, the trade-off between fund return and risk grade, and the dominance of lumpsum transactions over SIP by value even as SIP inflows remain a steady industry contributor.

On a personal level, this internship taught me that the most time-consuming part of a dashboard project is rarely the visualization itself — it is getting the data clean, consistent and correctly modeled beforehand. I also came away with a much better sense of how to translate a business question into a specific chart or metric rather than defaulting to whatever visualization looks most impressive. I believe the dashboard delivers real business value to Bluestock Fintech by giving stakeholders a faster, more reliable way to understand fund performance and investor activity, and I am glad to have been able to contribute it as part of my internship.

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